GLANSIS ORGANISM RISK ASSESSMENT

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SPECIES INFORMATION

Scientific Name: *Cherax destructor* (Clark, 1936)

Common Name: Yabby

SECTION A: INTRODUCTION POTENTIAL RESULTS

Dispersal: Unlikely
Hitchhiking/Fouling: Unlikely
Unauthorized intentional release: Moderate
Stocking/Planting/Escape from recreational culture: Unlikely
Escape from commercial culture: Unlikely
Shipping: Unlikely

***Cherax destructor* has a moderate probability of introduction to the Great Lakes (Confidence level: High).**

Potential pathway(s) of introduction: Unauthorized intentional release

Comments:
Cherax destructor is known to be an invasive species in many areas outside of the United States, but it has not yet been introduced to the Great Lakes and has a moderate probability of introduction. The possible risk of introduction would be due to unauthorized intentional release.

POTENTIAL INTRODUCTION VIA DISPERSAL

A1) Does this species occur near waters (natural or artificial) connected to the Great Lakes basin* (e.g., streams, ponds, canals, or wetlands)? (*Great Lakes basin = below the ordinary high water mark, including connecting channels, wetlands, and waters ordinarily attached to the Lakes)

Yes, this species occurs near waters connected to the Great Lakes basin and is mobile or able to be transported by wind or water.	100
No, this species does not occur near waters connected to the Great Lakes basin and/or is not mobile or able to be transported by wind or water.	0 √
Unknown	U

- Cherax destructor *is naturally found in Australia; it has been found in Victoria, New South Wales, and Southern Queensland (USFWS 2014).*
- C. destructor *has been introduced to South Africa, Zambia, Spain, Italy, China, Switzerland, Austria, Netherlands, England, and Germany (Chucholl 2013, Gherardi and Allen 2022).*

A2) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 kilometers of the Great Lakes basin, and no barrier (e.g., electric barrier, dam) to dispersal is present.	Score x 1
This species occurs in waters within 20 kilometers of the Great Lakes basin, but dispersal to the basin is blocked; or, this species occurs in waters within 100 kilometers of the Great Lakes basin, and no barrier to dispersal is present.	Score x 0.75
This species occurs in waters within 100 kilometers of the Great Lakes basin, but dispersal to the basin is blocked.	Score x 0.5
This species occurs in waters >100 kilometers from the Great Lakes basin.	Score x 0.25
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA HITCHHIKING/FOULING

A3) Is this species likely to attach to or be otherwise transported by, or along with, recreational gear, boats, trailers, fauna (e.g., waterfowl, fish, insects), flora (e.g., aquatic plants), or other objects (e.g., packing materials), including as parasites or pathogens, entering the Great Lakes basin?

Yes, this species is known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	100
No, this species is not known to be able to adhere to certain surfaces or to be transported by other organisms entering the Great Lakes basin.	0 √
Unknown	U

- *The yabby is usually found buried in the sediment of muddy waters (Withnall 2000).*
- Cherax destructor *is currently a prohibited species in Michigan (Natural Resources and Environmental Protection Act 1994) and Ontario (Invasive species Act 2015) and listed as a restricted species in Minnesota (MN Administrative Code § 6216.0250), Illinois (IL Administrative Code § 805.20), and Ohio (OH Administrative Code § 1501:31-19), so intentional transport within the Great Lakes basin should be limited.*

- *At the federal level,* C. destructor *was added to the list of injurious crustaceans by the U.S. Fish and Wildlife Service in 2016 (50 CFR § 16.13).*

A4) What is the proximity of this species to the Great Lakes basin?

This species occurs in waters within 20 km of the Great Lakes basin.	Score x 1
This species occurs in waters within 100 km of the Great Lakes basin.	Score x 0.5
This species occurs in waters >100 km from the Great Lakes basin.	Score x 0.1
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA UNAUTHORIZED INTENTIONAL RELEASE

A5) Is this species sold at aquarium/pet/garden stores ("brick & mortar" or online), catalogs, biological supply companies, or live markets (e.g., purchased for human consumption, bait, ornamental, ethical, educational, or cultural reasons) and as a result may be released into the Great Lakes basin?

Yes, this species is available for purchase.	100 √
No, this species is rarely/never sold.	0
Unknown	U

- Cherax destructor *has a relatively high commercial value (Chucholl 2013).*
- Cherax destructor *is used as bait, and unused live bait is often being discarded directly into the water (Gherardi 2007).*
- *The yabby is considered a culinary delicacy in Europe and farmed in some countries (e.g. Italy).*
- *Yabby is an aquarium species that is easy to maintain; it can be purchased through e-commerce and is increasingly used in research.*
- *The potential for* C. destructor *specimens to be released and become established is extremely high, especially in temperate and warmer climates (Gherardi and Allen 2022).*
- *In 2009,* Cherax destructor *was one of the most popular species in aquarium trade in the Netherlands; popularity was found to have a direct link to chance of introduction/establishment (Chucholl 2013).*

A6) How easily is this species obtained within the Great Lakes region (states/provinces)?

This species is widely popular, frequently sold, and/or easily obtained within the Great Lakes region.	Score x 1
This species is widely popular, and although trade, sale, and/or possession of this species is prohibited, it is frequently sold on the black market within the Great Lakes region.	Score x 0.5 √
This species is not very popular or is not easily obtained within the Great Lakes region.	Score x 0.1
Unknown	U

- Cherax destructor *is accessible in the United States in areas outside the Great Lakes region. There are none currently found in the Great Lakes region.*
- *Shipping for aquarium trade between the United States and Europe decreased due to crayfish plague, which originated from the crayfish brought from the US; the plague was first recorded in 1859 and removed up to 95% of the indigenous European crayfish species (Chucholl 2013).*
- Cherax destructor *is currently a restricted species in Minnesota, Michigan, Illinois, and Ohio. It is also listed as an injurious species by the U.S. Fish and Wildlife Service and prohibited in Ontario, Canada.*

- Cherax destructor*is available for sale online in the United States and could easily be purchased by someone in the Great Lakes region (e.g., Ebay).*
- Cherax destructor *can also be purchased from outside the United States and shipped (e.g., Blue Yabby Australia).*
- *Sixteen online shops offer 37 different nonindigenous crayfish species, including* C. destructor*, and on a scale from very rare to very common, the yabby was considered a common species to be found in aquariums (Chucholl 2013). However, the availability of* C. destructor *in the pet market may have declined as recent research reported its availability as rare with an expensive price point compared to other non-indigenous crayfish species (Vodovskoy et al. 2017). This species is still available for purchase and the pet trade remains a potential pathway for introduction.*

POTENTIAL INTRODUCTION VIA STOCKING/PLANTING OR ESCAPE FROM RECREATIONAL CULTURE

A7) Is this species being stocked/planted to natural waters or outdoor water gardens around the Great Lakes region?

Yes, this species is being stocked/planted and/or has ornamental, cultural, medicinal, environmental (e.g., biocontrol, erosion control), scientific, or recreational value in the Great Lakes region.	100
No, this species cannot be stocked/planted or there is not enough interest to do so in the Great Lakes region.	0 √
Unknown	U

- *This review found no record of* C. destructor *being stocked in close proximity to the Great Lakes.*
- Cherax destructor *is currently a restricted species in Minnesota, Michigan, Illinois, and Ohio. It is also listed as an injurious species by the U.S. Fish and Wildlife Service and prohibited in Ontario, Canada. Therefore, it is very unlikely to be stocked or planted in natural waters.*

A8) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is authorized and/or is occurring directly in the Great Lakes.	Score x 1
This activity is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is likely to occur within 20 km of the Great Lakes basin because of its popularity/value and there are no widespread regulations against stocking/planting.	Score x 0.5
This activity is occurring in waters >20 km from the Great Lakes basin, or despite federal or state regulations in more than half the basin (> 5 states/provinces), this activity may occur within 20 km of the basin because of the species' popularity/value.	Score x 0.25
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA ESCAPE FROM COMMERCIAL CULTURE

A9) Is this species known to be commercially cultured in or transported through the Great Lakes region?

Yes, this species is being commercially cultured in or transported through the Great Lakes region.	100

No, this species is not commercially cultured in or transported through the Great Lakes region.	0 √
Unknown	U

- Cherax destructor *is not known to be transported through the Great Lakes region, but its transport has been increasing globally, increasing the future risk of introduction.*
- Cherax destructor *is currently a prohibited species in Michigan (Natural Resources and Environmental Protection Act 1994) and Ontario (Invasive species Act 2015) and listed as a restricted species in Minnesota (MN Administrative Code § 6216.0250), Illinois (IL Administrative Code § 805.20), and Ohio (OH Administrative Code § 1501:31-19), so intentional transport within the Great Lakes basin should be limited.*
- C. destructor *was originally transported for aquaculture and food, but more recently, it has gained popularity as a pet (Chucholl 2013).*
- *Out of the 600 crayfish species that are found worldwide, the yabby is one of the few with commercial value (Harlioğlu and Harlioğlu 2006).*
- *Since the 1970s,* C. destructor *was intentionally introduced into different areas, specifically in Africa and China for aquacultural purposes (Harlioğlu and Harlioğlu 2006).*

A10) What is the nature and proximity of this activity to the Great Lakes basin?

This activity is unregulated or minimally regulated and is occurring directly in the Great Lakes.	Score x 1
This activity is unregulated or minimally regulated and is occurring in Great Lakes tributaries or connecting waters, or within 20 km of the Great Lakes basin.	Score x 0.75
This activity is strictly regulated but occurs directly in the Great Lakes, and/or this activity involves transport of live organisms on/across the Great Lakes.	Score x 0.5
This activity is strictly regulated but occurs in Great Lakes tributaries, connecting waters, or within 20 km of the Great Lakes basin, and/or this activity involves transport of live organisms within 20 km of the Great Lakes basin.	Score x 0.25
This activity occurs >20 km from the Great Lakes basin and typically does not involve transport of live organisms closer to the basin.	Score x 0.1
Unknown	U

- *N/A*

POTENTIAL INTRODUCTION VIA SHIPPING

A11) Is this species capable of surviving adverse environments (i.e. extreme temperatures, absence of light, low oxygen levels) and partial-to-complete ballast water exchange (BWE) (e.g., is euryhaline, buries in sediment, produces resistant resting stages, has other attributes or behaviors facilitating survival under these conditions)?

Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water.	100
Yes, this species is able to survive in ballast tank environments for weeks at a time and is able to survive BWE by burial in ballast sediment.	80 √
Yes, this species is able to survive in ballast tank environments for weeks at a time and may be suspended in ballast water, but this species is not able to survive BWE.	60

No, but this species is capable of fouling transoceanic ship structures (e.g., hull, chains, chain locker) while in its active or resting stage.	40
No, this species is not able to survive adverse environments, does not foul transoceanic ship structures, or is unlikely to be taken up with ballast.	0
Unknown	U

- Cherax destructor *has a large tolerance to changing environments (such as temperature, water levels, oxygen levels, and salinity); it can occupy different habitats for a short period of time and can quickly colonize in new environments (USFWS 2014).*
- *The yabby has a temperature tolerance of 1-35°C. It stops growing when salinity levels reach 8 ppt and death occurs at 16 ppt. Oxygen levels reaching less than 1 mg/L and a pH range of 7 to 9 can be tolerated (USFWS 2014).*
- *A field study of the yabby reported a chronic salinity tolerance of approximately 20 ppt (Beatty 2005).*
- C. destructor *is unlikely to survive for weeks at a time in ballast water; research shows that adults exposed to a salinity of 31.6 ppt died within 96 hours (Mills and Geddes 1980).*
- C. destructor *is a hyporheic dweller that constructs vertical burrows (Kouba et al. 2016). The yabby has the potential to survive ballast water exchange by burial in ballast sediment.*

A12) Does this species occur in waters from which shipping traffic to the Great Lakes originates?

Yes, and this species has been observed in the ballast of or fouling ships entering the Great Lakes.	Score x 1
Yes, and this species has been observed in ports that have direct trade connections with the Great Lakes (e.g., Baltic Sea).	Score x 0. 5
Yes, but this species has neither been observed in ballast/fouling ships entering the Great Lakes nor in ports in direct trade with the Great Lakes.	Score x 0.1
No, this species does not occur in waters from which shipping traffic to the Great Lakes originates.	Score x 0 √
Unknown	U

- *There is no evidence at this time that there is direct shipping traffic to the Great Lakes region from the Australian, Chinese, and European waters where the yabby is present.*

Potential Vector Ranking and Points

Vector	**Raw Points Scored**	**Proximity Multiplier**	**Total Points Scored**	**Probability of Introduction**
Dispersal: Natural dispersal through waterbody connections or wind	0	x	0	Unlikely
Hitchhiking/Fouling: Transport via recreational gear, boats, trailers, mobile fauna, stocked/planted organisms, packing materials, host organisms, etc.	0	x	0	Unlikely
Release: Unauthorized intentional release of organisms in trade (e.g., aquaria, water gardens, live food)	100	x 0.5	50	Moderate
Stocking/Planting/Escape from recreational culture: Intentional authorized or unauthorized	0	x	0	Unlikely

introduction to natural waters in the Great Lakes *OR* Accidental introduction to Great Lakes by escape from recreational culture (e.g., water gardens)					
Escape from commercial culture: Accidental introduction to Great Lakes by escape from commercial culture (e.g., aquaculture)		0	x	0	Unlikely
Trans-oceanic shipping: Ballast (BOB) or no-ballast-on-board (NOBOB) water exchange/discharge, sediment discharge, hull fouling		80	x 0	0	Unlikely
Total Unknowns (U)	0	Confidence Level		High	

Qualitative Statements for GLANSIS Fact Sheet:

Cherax destructor has a moderate probability of introduction to the Great Lakes (Confidence level: High).

Potential pathway(s) of introduction: Unauthorized intentional release

Scoring	
Points (per vector)	**Probability for Introduction**
80-100	High
40-79	Moderate
1-39	Low
0	Unlikely
# of Unknowns (overall)	**Confidence Level**
0	High
1-2	Moderate
3-5	Low
>5	Very low

SECTION B: RISK OF ESTABLISHMENT RESULTS

INVASIVE BIOLOGICAL/ECOLOGICAL ATTRIBUTES

B1) How would the physiological tolerance of this species (survival in varying temperature, salinity, oxygen, and nutrient levels) be described?

This species has broad physiological tolerance. It has been reported to survive in wide ranges of temperature (0°C-30°C), salinity (0-16 parts per thousand), oxygen (0-saturated), AND nutrient (oligotrophic-eutrophic) levels.	9
This species has somewhat broad physiological tolerance. It has been reported to survive in a wide range of temperature, salinity, oxygen, OR nutrient levels. Tolerance to other factors is narrower, unknown, or unreported.	6
This species has narrow physiological tolerance. It has been reported to survive in limited ranges of temperature, salinity, oxygen, and nutrient levels.	3
Unknown	U
	9

- Cherax destructor *is known to have a high tolerance to a variety of environments.*
- *This species is generally found where there are high oxygen levels and a good source of vegetation (Withnall 2000).*
- *This species' ideal temperature range is 20-25°C, but they are able to survive in 1-35°C while growth stops below 16°C and above 35°C (Withnall 2000).*
- *High salinity levels are tolerated by the yabby, but only for a period of time. They can survive in sea water for up to 48 hours, but after this stress on their systems, they stop growing (Withnall 2000). A field study of the yabby reported a chronic salinity tolerance of approximately 20 ppt (Beatty 2005).*
- *Yabby can tolerate low oxygen levels, but after long periods of time, their growth will stop (Withnall 2000).*
- *The yabby prefers muddy waters opposed to clear water environments (Withnall 2000), and this species survives dry periods by burrowing in the mud (Geddes and Smallridge 1993).*
- *They are polytrophic and have high fecundity, which increases their ability to survive (Geddes and Smallridge 1993).*
- *The eggs of* Cherax destructor *were tested and were able to tolerate stress from changes in the environment, but they were not as harvestable. The conditions they tested were temperature, oxygen levels, salinity, and light (Geddes and Smallridge 1993).*

B2) How likely is it that any life stage of this species can overwinter in the Great Lakes (survive extremely low levels of oxygen, light, and temperature)?

Likely (This species is able to tolerate temperatures under 5°C and oxygen levels ≤0.5 mg/L)	9
Somewhat likely (This species is able to tolerate some of these conditions OR has adapted behaviorally to avoid them)	6
Somewhat unlikely (This species is able to tolerate conditions close to those specified, but it is not known as an overwintering species)	3
Unlikely	0
Unknown	U
	6

- Cherax destructor *has a high climate match with the United States, particularly in the East, Texas, and Northwest. A climate match of 0.103 and higher is considered a high match, a score greater than 0.005 and less than 0.103 is considered a medium match, while a score less than or equal to 0.005 is categorized as a low match. The yabby has a match of 0.388. However, the climate match for the Great Lakes region is either medium or low. The climate match for New York and Indiana is medium while the match for Pennsylvania, Ohio, Michigan, Minnesota, and Wisconsin is low (USFWS 2019).*
- *The extreme temperatures near the Great Lakes may affect the growth of the yabby during some seasons, but their high proliferation would help them survive. They have two spawning events per season and produce approximately 1000 eggs per spawn. Reproduction is linked to water temperature, so the spring and summer would be ideal seasons (Withnall 2000).*
- *Through a climate suitability analysis, the yabby was identified as one species with the potential to overwinter and possibly establish viable populations in the Caspian Sea region, suggesting a similar potential in the Great Lakes (Vodovsky et al. 2017).*

B3) If this species is a heterotroph, how would the flexibility of its diet be described?

This species is a dietary generalist with a broad, assorted, AND flexible diet.	9
This species is moderately a dietary generalist with a broad, assorted, OR flexible diet.	6
This species is a dietary specialist with a limited and inflexible diet.	3
This species is an autotroph.	0
Unknown	U
	9

- Cherax destructor *is known to be an omnivorous species that feeds on plants, detritus, and some arthropods (Gherardi and Allen 2022).*
- *The yabby primarily consumes high protein prey (i.e. fish), with the mosquitofish* Gambusia holbrookii *constituting a majority of their summer diet in Australia outside of their native range. When this prey becomes limited during winter, however, they have demonstrated the ability to switch trophic positions and their diet shifts to plants/detritus (Beatty 2005).*
- C. destructor *may have cannibalistic tendencies in areas where there is an overabundance of the species and low levels of natural food sources (Gherardi and Allen 2022).*

B4) How likely is this species to outcompete species in the Great Lakes for available resources?

Likely (This species is known to have superior competitive abilities and has a history of outcompeting other species, AND/OR available literature predicts it might outcompete native species in the Great Lakes)	9
Somewhat likely (This species is known to have superior competitive abilities, but there are few reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	6
Somewhat unlikely (This species has average competitive abilities, and there are no reported cases of this species outcompeting another and no predictions regarding species in the Great Lakes)	3
Unlikely (This species is known as a poor competitor that thrives only in environments with low biodiversity, AND/OR available literature predicts it might be outcompeted by a species in the Great Lakes)	0
Unknown	U
	9

- Cherax destructor*'s adaptive diet suggests it may compete with smaller native freshwater crayfish for resources, especially in small, unproductive lotic and lentic ecosystems (Beatty 2005).*
- *The yabby has shown antagonistic behavior to get access to limited resources. In a trial competition between the yabby and the marron,* Cherax tenuimanus*, the yabby won. They concluded that its aggression led to success (Gherardi 2007).*
- *In contest experiments between* C. destructor *and the giant spiny crayfish* Euastacus spinifer, the yabby had a competitive advantage regardless of water temperature, habitat complexity, or residency status (Cerato et al. 2019, O'Hea et al. 2022).
- *An observational field study using an underwater video camera documented competitive interactions between the yabby and the endangered Fitzroy Falls spiny crayfish* Euastacus dharawalus. *The yabby displayed increased aggression and competitive dominance (O'Hea et al. 2022).*
- *The success of another invasive crayfish species, rusty crayfish or* Faxonius rusticus *could give some insight to how the yabby would compete in the Great Lakes region. The rusty crayfish is native to the United States but has invaded Minnesota, Wisconsin, Michigan, Illinois, and other states. Its spread has caused a reduction in the native crayfish population, aquatic plants, and fish populations (Gunderson 2004). The rusty crayfish is similar to the yabby because it is a larger crayfish species with an aggressive behavior and an opportunistic diet (Gunderson 2004).*

B5) How would the fecundity of this species be described relative to other species in the same taxonomic Class?

Very high	9
High	6
Moderate	3
Low	0
Unknown	U
	3

- *When considering other crayfish species, the yabby is mostly considered a r-selected species which has characteristics of short life cycles, high fecundity, and fast growth. These are also characteristics of good invaders (Gherardi 2007).*
- *The fecundity of the yabby is high and similar to other invasive crayfish species, such as the marbled crayfish* Procambarus virginalis *and red swamp crayfish* Faxonius rusticus *(Geddes and Smallridge 1993; Kouba et al. 2021).*

B6) How likely are this species' reproductive strategy and habits to aid establishment in new environments, particularly the Great Lakes (e.g., parthenogenesis/self-crossing, self fertility, vegetative fragmentation)?

Likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, AND available literature predicts establishment in the Great Lakes based on these attributes)	9
Somewhat likely (The reproductive strategy or habits of this species are known to aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	6
Somewhat unlikely (The reproductive strategy or habits of this species could potentially aid establishment in new environments, but there is no literature available regarding establishment in the Great Lakes based on these attributes)	3

Unlikely (The reproductive strategy or habits of this species are not known to aid establishment in new environments)	0
Unknown	U
	3

- *Reproductive strategies of* Cherax destructor *include multiple spawning events each year, high spawning frequency, a fast growth rate, and attainment of sexual maturity early in its life span (Beatty 2005).*
- *The reproductive habits of this species is affected by increased day length and water temperature. The eggs rest on the females swimming legs while waiting to hatch (USFWS 2014).*
- C. destructor *is capable of reproducing five times a year in suitable conditions and reaches sexual maturity at less than one year old. Clutch size is typically 300 to 500 eggs but instances of large females carrying 1400 eggs have been reported (Kouba et al. 2021).*
- *Females carrying eggs provide parental care by ensuring a well oxygenated environment and keeping eggs clean and free of foreign particles (Withnall 2000).*
- *At the time of this review, no literature specific to* C. destructor *establishment in the Great Lakes is available.*

ENVIRONMENTAL COMPATIBILITY

B7) How similar are the climatic conditions (e.g., air temperature, precipitation, seasonality) in the native and introduced ranges of this species to those in the Great Lakes region?

Very similar (The climatic conditions are practically identical to those of the Great Lakes region)	9
Similar (Many of the climatic conditions are similar to those of the Great Lakes region)	6
Somewhat similar (Few of the climatic conditions are similar to those of the Great Lakes region)	3
Not similar	0
Unknown	U
	5

- Cherax destructor *has a high climate match with the United States, particularly in the East, Texas, and Northwest. A climate match of 0.103 and higher is considered a high match, a score greater than 0.005 and less than 0.103 is considered a medium match, while a score less than or equal to 0.005 is categorized as a low match. The yabby has a match of 0.388. However, the climate match for the Great Lakes region is either medium or low. The climate match for New York and Indiana is medium while the match for Pennsylvania, Ohio, Michigan, Minnesota, and Wisconsin is low (USFWS 2019).*
- *Through a climate suitability analysis, the yabby was identified as one species with the potential to overwinter and possibly establish viable populations in the Caspian Sea region, suggesting a similar potential in the Great Lakes (Vodovskoy et al. 2017).*
- *When comparing climate conditions of a native region (New South Wales, Australia) to a potential new region in the Great Lakes (Detroit, Michigan), there are some similarities and differences. The temperature range in Australia is less drastic than in Michigan and average minimum temperature is higher. In Australia, the average maximum temperature ranges from 63°F to 79°F, and the average minimum temperature ranges from 45°F to 66°F. In Michigan, the average maximum temperature ranges from 29°F to 84°F, and the average minimum temperature ranges from 15°F to 63°F. (About Australia 2023, Weather Spark 2023). The Great Lakes region frequently experiences lower temperatures than what is observed in Australia during the winter months and water temperatures often fall below the optimum range (20-28 °C, Gherardi and Allen 2022) for* C. destructor *growth.*

- *The two locations have similar rainfall measurements but during different months. The most rainfall occurs from January to June in Australia and from May to September in Michigan (About Australia 2023, Weather Spark 2023).*

B8) How similar are other abiotic factors that are relevant to the establishment success of this species (e.g., pollution, water temperature, salinity, pH, nutrient levels, currents) in the native and introduced ranges to those in the Great Lakes?

Very similar (These factors are practically identical to those of the Great Lakes region)	9
Similar (Many of these factors are similar to those of the Great Lakes region)	6
Somewhat similar (Few of these factors are similar to those of the Great Lakes region)	3
Not similar	0
Unknown	U
	7

- Cherax destructor *is adapted to a wide range of water temperatures, between 1°C and 35°C. It does not grow at water temperatures below 15°C and falls into a state of partial hibernation (i.e. metabolism and feeding cease) when water temperature drops below 16°C (Withnall 2000).*
- C. destructor *tolerates high salinities, with growth ceasing at 8 ppt (approximately equal to 25% seawater) and mortalities starting to occur at 16 ppt (Mills and Geddes 1980). It tolerates oxygen concentration <1 mg L^{-1}, being able to survive for a short time at 0 mg L^{-1} oxygen (Morrissy and Cassells 1990).*
- *Yabbies are commonly found on muddy or silted bottoms and are rarely found in clear water habitats; they seem to prefer water with moderate levels of turbidity. Possibly, muddy waters afford some protection from predators such as fish and birds, giving the yabby a better chance of survival (Withnall 2000).*
- *The Great Lakes region frequently experiences lower temperatures than what is observed in Australia during the winter months and water temperatures often fall below the optimum range (22-30 °C, Gherardi and Allen 2022) for* C. destructor *growth.*

B9) How abundant are habitats suitable for the survival, development, and reproduction of this species in the Great Lakes area (e.g., those with adequate depth, substrate, light, temperature, oxygen)?

Abundant (Suitable habitats can be easily found and readily available)	9
Somewhat abundant (Suitable habitats can be easily found but are in high demand by species already present)	6
Somewhat scarce (Suitable habitats can be found occasionally)	3
Scarce (Suitable habitats are rarely found)	0
Unknown	U
	9

- Cherax destructor*'s ideal habitat consists of high oxygen levels and vegetation; they are most likely to be found in swamps, streams, rivers, and dams (Withnall 2000). The Great Lakes region provides abundant habitat that is suitable for the survival, development, and reproduction of this species.*
- C. destructor *prefers muddy water and does not need to be fully covered by water unless mating; they require a certain water hardness that comes from dissolved calcium (NSW DPI 2017).*
- *The yabby has been the most successful and abundant crayfish in Australian freshwater; it has adapted to the cold waters in the mountains as well as the hot waters of central Australia (NSW DPI 2017).*

B10) How likely is this species to adapt to or to benefit from the predicted effects of climate change on the Great Lakes freshwater ecosystems (e.g., warmer water temperatures, shorter duration of ice cover, altered streamflow patterns, increased salinization)?

Likely (Most of the effects described above make the Great Lakes a better environment for establishment and spread of this species, OR this species could easily adapt to these changes due to its wide environmental tolerances)	9
Somewhat likely (Several of the effects described above could make the Great Lakes a better environment for establishment and spread of this species)	6
Somewhat unlikely (Few of the effects described above would make the Great Lakes a better environment for establishment and spread of this species)	3
Unlikely (Most of the effects described above would have no effect on establishment and spread of this species or would make the environment of the Great Lakes unsuitable)	0
Unknown	U
	9

- Cherax destructor *will likely benefit from the environmental differences and is known to have a high tolerance to a variety of environments (Gherardi and Allen 2022).*
- *The yabby's preferred climate is tropical or warm temperate, warmer water temperatures are expected to support establishment and reproductive success. Yabbies mate when water temperature is above 15 °C and optimal growth occurs at 22 to 30 °C (Gherardi and Allen 2022, McClain 2020).*
- *The yabby is a burrowing crayfish and would be expected to tolerate altered streamflow patterns and changing water levels. The burrowing behavior will support survival during dry periods; there are records of specimens surviving in burrows underneath lake beds that have been dry for years (Holdich 1999).*
- *The yabby is capable of tolerating increased salinization. A field study of the yabby reported a chronic salinity tolerance of approximately 20 ppt (Beatty 2005).*
- *The extreme temperatures near the Great Lakes may affect the growth of the yabby during some seasons, but their high proliferation would help them survive. They have two spawning events per season and produce approximately 1000 eggs per spawn. Reproduction is linked to water temperature, so the spring and summer would be ideal seasons (Withnall 2000).*

B11) How likely is this species to find an appropriate food source (prey or vegetation in the case of predators and herbivores, or sufficient light or nutrients in the case of autotrophs)?

Likely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are highly abundant and/or easily found)	9
Somewhat likely (Some nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is low to moderate)	6
Somewhat unlikely (Few nutritive food items—including species in the Great Lakes that may be considered potential food items—are abundant and/or search time is moderate to high)	3
Unlikely (All possible nutritive food items—including species in the Great Lakes that may be considered potential food items—are relatively scarce and/or search time is high)	0
Unknown	U
	9

- Cherax destructor *is an opportunistic feeder and would be able to find a food source in most environments (Withnall 2000).*

- C. destructor *is known to be an omnivorous species that feeds on plants, detritus, and some arthropods (Gherardi and Allen 2022).*
- *The yabby primarily consumes high protein prey (i.e. fish), with the mosquitofish* Gambusia holbrookii *constituting a majority of their summer diet in Australia outside of their native range. When this prey becomes limited during winter, however, they have demonstrated the ability to switch trophic positions and their diet shifts to plants/detritus (Beatty 2005).*

B12) Does this species require another species for critical stages in its life cycle such as growth (e.g., root symbionts), reproduction (e.g., pollinators, egg incubators), spread (e.g., seed dispersers), or transmission (e.g., vectors)?

Yes, and the critical species (or one that may provide a similar function) is common in the Great Lakes and can be easily found in environments suitable for the species being assessed; OR, No, there is no critical species required by the species being assessed	9
Yes, and the critical species (or one that may provide a similar function) is moderately abundant and relatively easily found in particular parts of the Great Lakes	6
Yes, and the critical species (or one that may provide a similar function) is relatively rare in the Great Lakes AND/OR can only be found occasionally in environments suitable for the species being assessed	3
Yes, and the critical species (or one that may provide a similar function) is not present in the Great Lakes but is likely to be introduced	0
Yes, but the critical species (or one that may provide a similar function) is not present in the Great Lakes and is not likely to be introduced	-80% total points (at end)
Unknown	U
	9

- *At the time of this assessment,* C. destructor *does not have any known organisms that it depends on.*

B13) How likely is the establishment of this species to be aided by the establishment and spread of another species already in the Great Lakes?

Likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes, AND available literature predicts this previous invader might promote the establishment of this species, AND/OR there have been cases reported of this species aiding the establishment of this species in other areas)	9
Somewhat likely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established and spread in the Great Lakes)	6
Somewhat unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species—a major host, food item, pollinator—has already established in the Great Lakes BUT it is still confined to a small area of the Lakes and the likelihood of encounter with this species assessed is hard to predict)	3
Unlikely (A non-indigenous species to the Great Lakes that facilitates the development of this species has not been established in the Great Lakes)	0
Unknown	U

	0

- *At the time of this review, there is not currently a species established in the Great Lakes region known to aid the establishment or spread of* Cherax destructor. *Additionally, there are no examples of a species facilitating the establishment of the yabby in its native or introduced ranges.*

B14) How likely is establishment of this species to be prevented by the herbivory, predation, or parasitism of a natural enemy that is already present in the Great Lakes and may preferentially target this species?

Likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is well documented in the literature AND this natural enemy is abundant and widespread in the Great Lakes)	-80% total points (at end)
Somewhat likely (The ability of the natural enemy to prevent the establishment of this species in introduced ranges or limiting populations of this species in native ranges is suggested in the literature OR this natural enemy has limited distribution in the Great Lakes)	-60% total points (at end)
Somewhat unlikely (There are few cases reported of such a natural enemy preventing the establishment of this species in introduced ranges or limiting populations of this species in native ranges OR this natural enemy has low abundance in the Great Lakes)	-10% total points (at end)
Unlikely (Such a natural enemy is particularly rare or is not present in the Great Lakes)	0
Unknown	U
	-10%

- Cherax destructor *prefers to settle in the mud or bottoms of muddy water which acts as protection and aids in survival (Withnall 2000).*
- *Predators are aquatic birds such as cormorants, herons and the ibis, and fish species such as Murray cod and callop. Carp compete with yabbies for food (Withnall 2000). All three bird species can be found in the Great Lakes region (MSG 2007).*

PROPAGULE PRESSURE

B15) On average, how large and frequent are inoculations (introduction events) from the potential vectors identified in Section A for this species? (What is the total number of individuals introduced?)

Frequent, large inocula	9
Frequent, moderate inocula	6
Frequent, small inocula OR infrequent, large inocula	3
Infrequent, small or moderate inocula	0
Unknown	U
	0

- *There have been no introductions of* Cherax destructor *in the Great Lakes region.*
- *While* C. destructor *is potentially being sold in the United States, there have been no reports of releases.*

- *There are increased restrictions in areas outside the United States to further prevent their release; in New South Wales Australia, it is illegal to release any crayfish into natural waters without government permission (NSW DPI 2017).*
- Cherax destructor *is currently a prohibited species in Michigan (Natural Resources and Environmental Protection Act 1994) and Ontario (Invasive species Act 2015) and listed as a restricted species in Minnesota (MN Administrative Code § 6216.0250), Illinois (IL Administrative Code § 805.20), and Ohio (OH Administrative Code § 1501:31-19),*

HISTORY OF INVASION AND SPREAD

B16) How extensively has this species established reproducing populations in areas outside its native range as a direct or indirect result of human activities?

Very extensively (many invasive populations of this species have been reported in areas widely distributed from the native range)	9
Extensively (some invasive populations of this species have been reported in areas widely distributed from the native range)	6
Somewhat extensively (few invasive populations of this species have been reported in areas widely distributed from the native range OR all invasive populations are in close proximity to each other)	3
Not extensively (no invasive populations of this species have been reported)	0
Unknown	U
	9

- Cherax destructor *is naturally found in Australia; it has been found in Victoria, New South Wales, and Southern Queensland (USFWS 2014).*
- *Its indigenous regions in Australia are South Australia, southern Northern territory, Victoria, and South Australia (Gherardi 2007).*
- *The yabby was introduced to Western Australia in 1932 by spreading through natural river systems (Gherardi 2007).*
- *This species has been introduced to South Africa, Zambia, Spain, Italy, China, Switzerland, Austria, Netherlands, England, and Germany (Chucholl 2013, Gherardi and Allen 2022).*

B17) How rapidly has this species spread by natural means or by human activities once introduced to other locations?

Rapidly (This species has a history of rapid spread in introduced ranges)	9
Somewhat rapidly (This species has a history of moderately rapid spread in introduced ranges)	6
Somewhat slowly (This species has a history of moderately slow spread in its introduced ranges)	3
Slowly (This species has a history of slow to no spread in its introduced ranges)	0
Unknown	U
	7

- Cherax destructor *was first moved out of Australia and introduced to Spain in 1983 for commercial reasons and was originally restricted to Spanish waters (Gherardi 2007).*

- *Since then, the yabby has moved throughout Europe and into Africa and was able to become established as a wild species (Gherardi 2007).*
- *As human interest in crayfish increased, so did the spread of the yabby (Gherardi 2007).*

B18) Are there any existing control measures in the Great Lakes set to prevent the establishment and/or spread of this species?

Yes, and they are likely to prevent establishment or spread of the species. (There are no reported cases of this species adapting or avoiding current measures. These measures are highly effective in preventing the establishment and spread of this species)	-90% total points (at end)
Yes, and they are moderately likely to prevent establishment or spread of the species. (There are few reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-50% total points (at end)
Yes, but they are unlikely to prevent establishment or spread of the species. (There are many reported cases of this species adapting or avoiding current measures used to control its establishment and spread)	-20% total points (at end)
No control methods have been set to prevent the establishment and/or spread of this species	0
Unknown	U
	-20%

- *Regulations have been set to prohibit species in the Great Lakes.* Cherax destructor *is currently a prohibited species in Michigan (Natural Resources and Environmental Protection Act 1994) and Ontario (Invasive species Act 2015) and listed as a restricted species in Minnesota (MN Administrative Code § 6216.0250), Illinois (IL Administrative Code § 805.20), and Ohio (OH Administrative Code § 1501:31-19).*
- *At the federal level,* C. destructor *was added to the list of injurious crustaceans by the U.S. Fish and Wildlife Service in 2016 (50 CFR § 16.13).*
- *C. destructor is a popular species in the aquarium trade (Chucholl 2013) and owned by hobbyists in the US. While current regulations restrict the importation of this species, it remains available through means outside of the regulatory scope (i.e. direct trade with other hobbyists, Ebay, etc.).*

Section B Scores				
Points	**Probability of Establishment**	**Total Points (pre-adjustment)**		103
>100	High	**Adjustments**		
		Critical species	A (103-0%)	103
51-99	Moderate	**Natural enemy**	B (103-10%)	93
		Control measures	C (93-20%)	74
0-50	Low	**Total Points (post-adjustment)**		**74**

		Probability of Establishment	Moderate
# of questions answered as "unable to determine"	**Confidence Level**		
0-1	High	**Total # of questions unknown**	**0**
2-5	Moderate		
6-9	Low	**Confidence Level**	**High**

Qualitative Statements for GLANSIS Fact Sheet:

Cherax destructor *has a moderate probability of establishment if introduced to the Great Lakes (Confidence level: High).*

Comments:
Since 1932, *Cherax destructor* has become a growing problem outside its native range. Its original invasion occurred within the Australian continent. Once people started developing an interest in this species, it spread across the globe to Europe and Africa. Wild populations have been established and pose a threat to native crayfish populations and other species.

The biology of the yabby has aided in its survival. The yabby has proven to have a high tolerance to many different environments and high fecundity. It also has an aggressive behavior that allows it to compete with other species for food. *Cherax destructor* does not have many predators in the Great Lakes region nor does it rely on certain species for survival. There are some regulatory measures set in place to prevent the further spread of *C. destructor*.

SECTION C: POTENTIAL IMPACT ASSESSMENT RESULTS

Environmental: High
Socio-Economic: Low
Beneficial: Moderate

Comments:

***Cherax destructor* has the potential for high environmental impact if introduced to the Great Lakes.**

Cherax destructor is not known to be toxic, parasitic, or poisonous. There have been cases where *C. destructor* has spread diseases and affected indigenous crayfish species. The crayfish plague in 1859 was caused by a parasite, *Aphanomyces astaci*, which was fatal to all European indigenous crayfish. *A. astaci* is thought to have originated from crayfish in the southeast US and the yabby is one of many carriers of this pathogen (Chucholl 2013; Martín-Torrijos et al. 2021). The yabby is a host for the following microsporidian parasites: *Vairimorpha cheracis*, *Thelohania montirivulorum*, and *Thelohania parastaci*. The *Theolohania* spp. cause thelohaniasis, commonly referred to as porcelain disease (Gherardi 2007, Pretto et al. 2018). Research has also identified the presence of PFAS in the edible tissue of the yabby, this species may bioaccumulate and transfer PFAS up the food web (Taylor 2018).

Cherax destructor is considered a keystone species that influences the aquatic food web through nutrient recycling (Beatty 2005). Due to its aggressive behavior and size, this species is capable of out-competing native species for resources. Ample research has emerged that supports the yabby's impact on native crayfish through behavioral interactions or competition for resources (Lopez et al. 2018, Cerato et al. 2019, King et al. 2022, O'Hea et al. 2022).

There is little or no evidence to support that *Cherax destructor* has the potential for significant socio-economic impacts if introduced to the Great Lakes.

It has not been reported that *Cherax destructor* poses a threat to human health or water quality. Burrowing activities of C. destructor can negatively impact commercial and recreational areas (Holdich 1999), but the effects have not been widespread or severe. This review found no evidence that this species negatively impacts economic sectors, recreational activities and associated tourism, or the aesthetic appeal of the areas it inhabits.

***Cherax destructor* has the potential for moderate beneficial impact if introduced to the Great Lakes.**

Cherax destructor has commercial value but its economic contribution is small. It has become a target species for aquaculture in several countries and is employed recreationally as bait but adds little value to local communities or tourism. This species has demonstrated utility in laboratory use and in research as a model organism (Gherardi and Allen 2022).

POTENTIAL ENVIRONMENTAL IMPACT

C1) Does the species pose some hazard or threat to the health of native species (e.g., it magnifies toxin levels; is poisonous; is a pathogen, parasite, or a vector of either)?

Yes, and it has impacted threatened/endangered species, resulted in the reduction or extinction of one or more native populations, affects multiple species, or is a reportable disease	6 √
Yes, but negative consequences have been small (e.g., limited number of infected individuals, limited pathogen transmissibility, mild effects on populations and ecosystems)	1
Not significantly	0
Unknown	U

- Cherax destructor *is not known to be toxic, parasitic, or poisonous.*
- *There have been cases where* C. destructor *has spread diseases and affected indigenous crayfish species.*
- *The crayfish plague in 1859 was caused by a parasite,* Aphanomyces astaci*, which was fatal to all European indigenous crayfish.* A. astaci *is thought to have originated from crayfish in the southeast US and the yabby is one of many carriers of this pathogen (Chucholl 2013; Martín-Torrijos et al. 2021). The crayfish plague* Aphanomyces astaci *is a reportable disease (USDA 2022).*
- *The yabby is also known to carry* Thelohania parastaci *which is a microsporidian disease that causes the destruction of striated and cardiac muscle tissue and eventually leads to death of the affected crayfish (Gherardi 2007).*
- C. destructor *is a known host for a systemic parvo-like virus. Little is known about the transmission, potential hosts, or viability of this virus but it could pose a threat to the health of other crayfish or crustaceans (Diggles 2011).*
- *The yabby is a host for the following microsporidian parasites:* Vairimorpha cheracis*,* Thelohania montirivulorum*, and* Thelohania parastaci*. The* Theolohania *spp. cause Thelohaniasis, commonly referred to as porcelain disease (Pretto et al. 2018).*
- *A field study in Australia observed low concentrations of two PFASs, perfluorooctane sulfonate and perfluorohexane sulfonate, in the yabby. The presence of these pollutants in the edible tissues of the yabby suggest this species may bioaccumulate and transfer PFASs up the food web (Taylor 2018).*

C2) Does it out-compete native species for available resources (e.g., habitat, food, nutrients, light)?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species or caused critical reduction, extinction, behavioral changes including modified spawning behavior) on one or more native populations	6 √
Yes, and it has caused some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population	1
Not significantly	0
Unknown	U

- C. destructor *has the ability to switch trophic positions, when an otherwise abundant, high protein food source (i.e. fish) becomes limited. Furthermore, the ability of* C. destructor *to switch from a diet of fish in summer to a predominantly herbaceous/detrital diet in winter suggests that it may compete for food resources with the other smaller native freshwater crayfishes in Australia (Beatty 2005).*
- *The yabby has shown antagonistic behavior to get access to limited resources. In a trial competition between the yabby and the marron,* Cherax tenuimanus*, the yabby won. They concluded that its aggression led to success (Gherardi 2007).*

- *In contest experiments between* C. destructor *and the giant spiny crayfish* Euastacus spinifer, *the yabby had a competitive advantage regardless of water temperature, habitat complexity, or residency status (Cerato et al. 2019, O'Hea et al. 2022).*
- *An observational field study using an underwater video camera documented competitive interactions between the yabby and the endangered Fitzroy Falls spiny crayfish* Euastacus dharawalus. *The yabby displayed increased aggression and competitive dominance (O'Hea et al. 2022).*
- *Competition experiments between the yabby and Australian redclaw crayfish* Cherax quadricarinatus *found that the yabby spent more time in possession of the food resource (King et al. 2022). A similar result was reported in contest experiments between the yabby and the endangered Fitzroy Falls spiny crayfish* Euastacus dharawalus. *The yabby spent significantly more time consuming food and may negatively impact the fitness of the Fitzroy Falls spiny crayfish through reduced access to resources (Lopez et al. 2018).*

C3) Does it alter predator-prey relationships?

Yes, and it has resulted in significant adverse effects (e.g., impacted threatened/endangered species, caused significant reduction or extinction of one or more native populations, creation of a dead end or any other significant alteration in the food web)	6
Yes, and it has resulted in some noticeable stress to (e.g., decrease in growth, survival, fecundity) or decline of at least one native population AND/OR Yes, and it has resulted in some alteration of the food web structure or processes, the effects of which have not been widespread or severe	1 √
Not significantly	0
Unknown	U

- *Research suggests that* C. destructor *is a trophic predator and functional omnivore. Considered a keystone species, the yabby greatly influences the aquatic food web through nutrient cycling (Beatty 2005).*

C4) Has it affected any native populations genetically (e.g., through hybridization, selective pressure, introgression)?

Yes, and it has caused a loss or alteration of genes that may be irreversible or has led to the decline of one or more native species (or added pressure to threatened/endangered species)	6
Yes, some genetic effects have been observed, but consequences have been limited to the individual level	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species affects native populations genetically.*

C5) Does it negatively affect water quality (e.g., increased turbidity or clarity, altered nutrient, oxygen, or other chemical levels/cycles)?

Yes, and it has had a widespread, long-term, or severe negative effect on water quality AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected water quality to some extent, but the alterations and resulting adverse effects have been limited or inconsistent (as compared with above statement)	1
Not significantly	0 √
Unknown	U

- C. destructor *is a hyporheic dweller that constructs vertical burrows to tolerate drought conditions (Kouba et al. 2016). The burrowing activity of the yabby may increase turbidity but the impact from this activity is not expected to be significant.*

C6) Does it alter physical components of the ecosystem in some way (e.g., facilitated erosion/siltation, altered hydrology, altered macrophyte/phytoplankton communities, physical or chemical changes to substrate)?

Yes, and it has had a widespread, long term, or severe negative effect on the physical ecosystem AND/OR Yes, and it has resulted in significant negative consequences for at least one native species	6
Yes, it has affected the physical ecosystem to some extent, but the alterations and resulting adverse effects have been mild	1
Not significantly	0 √
Unknown	U

- C. destructor *is a hyporheic dweller that constructs vertical burrows to tolerate drought conditions (Kouba et al. 2016). Burrowing activity of the yabby may increase erosion and destabilize river banks (OMNRF 2021), but no specific cases have been documented. The impact of yabby burrows on the physical ecosystem is not expected to be significant or severe.*

Environmental Impact Total	**13**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL SOCIO-ECONOMIC IMPACT

C7) Does the species pose some hazard or threat to human health (e.g., it magnifies toxin levels, is poisonous, a virus, bacteria, parasite, or a vector of one)?

Yes, significant effects on human health have already been observed	6
Yes, but negative consequences have not been widespread, long lasting, or severe	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species poses a hazard to human health.*

C8) Does it cause damage to infrastructure (e.g., water intakes, pipes, or any other industrial or recreational infrastructure)?

Yes, it is known to cause significant damage	6
Yes, but the costs have been small and are largely reparable or preventable	1 √
Not significantly	0
Unknown	U

- *The burrowing behaviour of* C. destructor *is a cause for concern for farmers. Yabbies are capable of digging very deep burrows which can be 50 cm to 2 meters deep depending on the species. Burrows are connected by access shafts to the water. In the event of the water drying up, the yabby is able to survive over summer in the burrows. Unfortunately, this behaviour may also destroy the integrity of dam walls causing problems for farmers (Withnall 2000).*
- *Burrowing activities of* C. destructor *can negatively impact recreational areas. For example, their burrows can affect residential lawns, golf courses, rivers, and lakes by destabilizing banks. (Holdich 1999).*
- *Yabby burrows can also negatively impact agricultural sectors. Their burrows can destabilize banks in rice plantations, levees, dams, and dykes (Holdich 1999).*

C9) Does it negatively affect water quality (i.e. in terms of being less suitable for human use)?

Yes, it has significantly affected water quality, and is costly or difficult to reverse	6
Yes, but the effects are negligible and/or easily reversed	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species negatively affects water quality.*

C10) Does it negatively affect any markets or economic sectors (e.g., commercial fisheries, aquaculture, agriculture)?

Yes, it has caused significant damage to one or more markets or economic sectors	6
Some damage to markets or sectors has been observed, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species causes harm to any markets or economic sectors.*

C11) Does it inhibit recreational activities and/or associated tourism (e.g., through frequent water closures, equipment damage, decline of recreational species)?

Yes, it has caused widespread, frequent, or otherwise expensive inhibition of recreation and tourism	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species inhibits recreation or tourism.*

C12) Does it diminish the perceived aesthetic or natural value of the areas it inhabits?

Yes, the species has received significant attention from the media/public, significantly diminished the natural or cultural character of the area, or significantly reduced the area's value for future generations	6
Yes, but negative consequences have been small	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species diminishes aesthetic or natural value of inhabited areas.*

Socio-Economic Impact Total	**1**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

POTENTIAL BENEFICIAL EFFECT

C13) Does it act as a biological control agent for aquatic weeds or other harmful nonindigenous organisms?

Yes, it has succeeded significantly as a control agent	6
Yes, it has had some success as a control agent, but may be inconsistent or lack a desired level of effectiveness	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species acts as a biological control agent.*

C14) Is it commercially valuable (e.g., for fisheries, aquaculture, agriculture, bait, ornamental trade)?

Yes, it is economically important to at least one of these industries	6
Yes, but its economic contribution is small	1 √
Not significantly	0
Unknown	U

- *In South Africa,* C. destructor *has become a target species for aquaculture and is commercially harvested (Holdich 1999).*
- *This is a popular species for aquaculture for purposes of eating and using in aquaria (NSW DPI 2017).*
- Cherax destructor *has a relatively high commercial value (Chucholl 2013).*

C15) Is it recreationally valuable (e.g., for sport or leisurely fishing, as a pet, or for any other personal activity)?

Yes, it is commonly employed recreationally and has some perceived value for local communities and/or tourism	6
It is sometimes employed recreationally, but adds little value to local communities or tourism	1 √
Not significantly	0
Unknown	U

- Cherax destructor *is used as bait, and unused live bait is often discarded directly into the water (Gherardi 2007).*
- C. destructor *is a popular species for recreational "catching" for purposes of eating and using as bait or in aquaria (NSW DPI 2017).*
- *In 2009,* C. destructor *was one of the most popular species in aquarium trade in the Netherlands (Chucholl 2013).*

C16) Does the species have some medicinal or research value (i.e. outside of research geared towards its control)?

Yes, it has significant medicinal or research value	6
It has some medicinal or research value, but is not of high priority OR It is potentially important to medicine or research and is currently being or scheduled to be studied	1 √

Not significantly	0
Unknown	U

- Cherax destructor *has research value, but it is not of high priority. This species has demonstrated utility in laboratory use and in research as a model organism (Gherardi and Allen 2022).*

C17) Does the species remove toxins or pollutants from the water or otherwise increase water quality?

Yes, it reduces water treatment costs or has a significant positive impact for the health of humans and/or native species	6
Yes, but positive impact for humans or native species is considered negligible	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species improves water quality.*

C18) Does the species have a positive ecological impact outside of biological control (e.g., increases the growth or reproduction rates of other species, fills an important gap in the food web, supports the survival of a species that is threatened, endangered species, or commercially valuable)?

Yes, it significantly contributes to the ecosystem in one or more of these ways	6
Yes, it provides some positive contribution to the ecosystem, but is not vital	1
Not significantly	0 √
Unknown	U

- *This review found no evidence that this species has any positive ecological impact outside of biological control.*

Beneficial Effect Total	**3**
Total Unknowns (U)	**0**

Scoring		
Score	# U's	**Impact**
>5	Any	**High**
2-5	Any	**Moderate**
0	0-1	**Low**
1	0	
0	≥2	**Unknown**
1	≥1	

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